# SQL Practical Lab Assignment

**Total Marks: 50**

## TechCart Logistics & Orders Data Analysis

**Database: MySQL 8.0+** | **Topic: DDL, DML, Aggregations & Group Operations** | **Target: Engineering Laboratory Practice**

**Business Problem Context:** You are working as a Junior Data Analyst at *TechCart India*. The logistics division needs SQL-based insights into product orders, device categories, fulfillment destinations, and settlement channels to streamline city deliveries and spot consumer spending patterns.

### Task 1: Database Setup & Selection

(4 Marks)

1. Write a query to create a database named `TechCart_Logistics`.
2. Select the `TechCart_Logistics` database for active querying.
3. List all available databases on the MySQL server instance.
4. Verify and display the currently selected active database schema using a built-in SQL function.

### Task 2: Schema Definition & Table Creation

(6 Marks)

Create a table named `Orders` according to the following specifications:

| Column Name | Data Type | Description & Constraints |
|---|---|---|
| `Order_ID` | `INT PRIMARY KEY` | Unique transaction identifier |
| `Customer_Name` | `VARCHAR(50)` | Full name of purchasing customer |
| `Device` | `VARCHAR(50)` | Commercial name of product/accessory |
| `Category` | `VARCHAR(30)` | Product line category (Audio, Wearables, Computing, Accessories) |
| `Quantity` | `INT` | Total units purchased in this order |
| `Unit_Price` | `DECIMAL(10,2)` | Selling price per single device unit (₹) |
| `City` | `VARCHAR(30)` | Delivery destination location |
| `Payment_Mode` | `VARCHAR(30)` | Settlement method (UPI, Credit Card, COD, Net Banking) |

### Task 3: Dataset Ingestion (20 Transaction Records)

(5 Marks)

Execute the batch insertion query to populate the orders ledger:

```
INSERT INTO Orders (Order_ID, Customer_Name, Device, Category, Quantity, Unit_Price, City, Payment_Mode) VALUES (101, 'Aarav Sharma', 'Smartwatch Pro', 'Wearables', 2, 4500.00, 'Ahilyanagar', 'UPI'), (102, 'Pooja Patil', 'Wireless Earbuds', 'Audio', 5, 1800.00, 'Pune', 'Credit Card'), (103, 'Rohan Deshmukh', 'Bluetooth Speaker', 'Audio', 1, 3200.00, 'Mumbai', 'COD'), (104, 'Ananya Joshi', 'Tablet 10-inch', 'Computing', 1, 22000.00, 'Nashik', 'Net Banking'), (105, 'Kunal Shinde', 'Fitness Band', 'Wearables', 4, 2100.00, 'Pune', 'UPI'), (106, 'Snehal Kadam', 'Wireless Earbuds', 'Audio', 3, 1800.00, 'Ahilyanagar', 'UPI'), (107, 'Vikas More', 'Mechanical Keyboard', 'Computing', 2, 3500.00, 'Chhatrapati Sambhajinagar', 'Credit Card'), (108, 'Neha Thorat', 'Smartwatch Pro', 'Wearables', 1, 4500.00, 'Mumbai', 'COD'), (109, 'Tanmay Pawar', 'Gaming Headset', 'Audio', 2, 4200.00, 'Pune', 'UPI'), (110, 'Priya Jagtap', 'Tablet 10-inch', 'Computing', 2, 21500.00, 'Mumbai', 'Credit Card'), (111, 'Saurabh Gaikwad', 'Wireless Mouse', 'Computing', 6, 750.00, 'Nashik', 'UPI'), (112, 'Aditi Kulkarni', 'Bluetooth Speaker', 'Audio', 3, 3100.00, 'Pune', 'COD'), (113, 'Nitin Chavan', 'Power Bank 20k', 'Accessories', 5, 1600.00, 'Chhatrapati Sambhajinagar', 'UPI'), (114, 'Swati Salunke', 'Fitness Band', 'Wearables', 2, 2100.00, 'Ahilyanagar', 'Credit Card'), (115, 'Akash Kale', 'Smartwatch Pro', 'Wearables', 3, 4400.00, 'Pune', 'Net Banking'), (116, 'Rutuja Bhise', 'Gaming Headset', 'Audio', 1, 4200.00, 'Nashik', 'COD'), (117, 'Omkar Wagh', 'Mechanical Keyboard', 'Computing', 3, 3400.00, 'Mumbai', 'UPI'), (118, 'Shraddha Gite', 'Power Bank 20k', 'Accessories', 4, 1550.00, 'Pune', 'UPI'), (119, 'Pratik Sonawane', 'Wireless Mouse', 'Computing', 8, 700.00, 'Ahilyanagar', 'COD'), (120, 'Divya Nalawade', 'Bluetooth Speaker', 'Audio', 2, 3200.00, 'Chhatrapati Sambhajinagar', 'Credit Card');
```

### Task 4: Filtering & Query Operations (6 Marks)

1. Display all records from the `Orders` table.
2. Display only `Customer_Name`, `Device`, and `Unit_Price`.
3. Display all orders destined for delivery in Pune.
4. Display all products categorized under `Audio`.
5. Display orders where `Unit_Price` is greater than ₹4,000.
6. Display orders where `Quantity` ordered is ≥ 3.
7. Display all transactions completed using `UPI`.
8. Display orders where `Unit_Price` is between ₹1,500 and ₹5,000.
9. Display orders placed from `Ahilyanagar` or `Nashik`.
10. Display devices whose names contain the substring `'Speaker'`.
11. Display all orders sorted in descending order of `Unit_Price`.
12. Display the top 4 most expensive orders by unit price.

### Task 5: Update Operations (4 Marks)

1. Update the delivery city of `Order_ID` 103 to Pune.
2. Increase unit price of all `'Wireless Mouse'` items to ₹800.
3. Update the payment mode of `Order_ID` 114 to `'UPI'`.
4. Apply a 5% discount to all products in `Computing` category.
5. Change category of `'Gaming Headset'` to `'Gaming Gear'`.

### Task 6: Delete Operations (3 Marks)

1. Delete the record where `Order_ID` is 116.
2. Delete all orders that used `'COD'` as payment method.
3. Delete orders where ordered `Quantity` is strictly less than 2.
4. Delete all records belonging to `'Accessories'` category.

### Task 7: Aggregate Functions (5 Marks)

1. `COUNT()`: Find the total number of orders placed.
2. `SUM()`: Calculate total units of devices ordered.
3. `AVG()`: Compute the average unit price across items.
4. `MAX()` & `MIN()`: Find highest and lowest unit prices.
5. `COUNT()`: Count how many orders were paid via `'UPI'`.
6. `SUM()`: Find total quantity delivered to

### Task 8: Grouping Data (GROUP BY) (6 Marks)

1. Find the total quantity sold in each product category.
2. Count the total number of orders placed from each city.
3. Calculate the average unit price per product category.
4. Find total number of items sold per payment mode.
5. Find maximum unit price recorded within each category.
6. Find minimum unit price of devices ordered in each city.
7. Calculate total units sold for each individual device name.
8. Find the total number of orders by payment mode.
9. Find total units sold grouped by both `City` and `Category`.
10. Count orders grouped by both `Device` and `Payment_Mode`.

### Task 9: Group Filtering (HAVING Clause) (6 Marks)

1. Display categories where total quantity sold exceeds 10 units.
2. Display cities that have received more than 4 orders.
3. Display payment modes used for more than 4 orders.
4. Display categories whose average unit price is > ₹3,000.
5. Display devices where total quantity ordered is ≥ 6 units.
6. Display cities where total quantity delivered is > 8 units.
7. Display payment modes whose total order quantity > 12 units.
8. Display categories featuring 3 or more distinct device models.
9. Display cities where average unit price of goods > ₹3,500.
10. Display categories whose maximum unit price exceeds ₹15,000.

### Task 10: Business Intelligence & Insights (10 Marks)

1. Which specific device carries the absolute highest unit price?
2. Which category generated the highest total volume of units?
3. Which delivery city registered the highest number of orders?
4. What is the most frequently chosen payment mode?